

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.6695989741090054, median 1.1932688208334554, std: 1.547072328760244
Reprojection error (cam1): mean 1.5979456840667678, median 1.169856751340713, std: 1.526831471169389
Gyroscope error (imu0): mean 4.008194965741186, median 3.211227621102286, std: 3.2106370193102287
Accelerometer error (imu0): mean 3.3769699123694945, median 2.7948369633452903, std: 2.8112995622568606

Residuals

Reprojection error (cam0) [px]: mean 1.6695989741090054, median 1.1932688208334554, std: 1.547072328760244
Reprojection error (cam1) [px]: mean 1.5979456840667678, median 1.169856751340713, std: 1.526831471169389
Gyroscope error (imu0) [rad/s]: mean 0.003106307137290338, median 0.0024886661861892095, std: 0.0024882084762779784
Accelerometer error (imu0) [m/s^2]: mean 0.02478617701245588, median 0.020513455995177794, std: 0.020634287658265058

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.00354472 -0.9999827 -0.00469494 0.04802988]
[-0.0008835 0.00469184 -0.9999886 0.01392703]
[0.99999333 0.00354882 -0.00086686 -0.01240217]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.00354472 -0.0008835 0.99999333 0.01224414]
[-0.9999827 0.00469184 0.00354882 0.04800772]
[-0.00469494 -0.9999886 -0.00086686 0.01414162]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.12947286717492387

Transformation (cam1):

T_ci: (imu0 to cam1):

```
[[ 0.00634791 -0.99996855 -0.0047534 -0.10182986]
 [ 0.01835073 0.00486918 -0.99981975 0.01356327]
 [ 0.99981146 0.00625954 0.01838106 -0.00677679]
 [ 0.         0.         0.         1.         ]]
```

T_ic: (cam1 to imu0):

```
[[ 0.00634791 0.01835073 0.99981146 0.00717302]
 [-0.99996855 0.00486918 0.00625954 -0.10185028]
 [-0.0047534 -0.99981975 0.01838106 0.01320135]
 [ 0.         0.         0.         1.         ]]
```

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.11983904537408437

Baselines:

Baseline (cam0 to cam1):

```
[[ 0.99999607 0.00005605 0.00280328 -0.14982557]
 [-0.00010996 0.99981499 0.01923458 -0.00011736]
 [-0.00280168 -0.01923482 0.99981107 0.00602549]
 [ 0.         0.         0.         1.         ]]
```

baseline norm: 0.14994673085519228 [m]

Gravity vector in target coords: [m/s^2]
[-0.06822385 -0.09504532 -9.80585207]

Calibration configuration

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cam0

Camera model: pinhole
Focal length: [595.5131125646116, 595.4347842307814]
Principal point: [937.1438588828621, 459.52906005794046]
Distortion model: radtan
Distortion coefficients: [-0.006053561997736177, -0.010116448442781898, -0.0007332457524041055, -0.0017822070415447345]
Type: aprilgrid
Tags:
Rows: 3
Cols: 4
Size: 0.046 [m]
Spacing 0.0138 [m]

cam1

Camera model: pinhole
Focal length: [604.5403230213659, 604.3848715029433]
Principal point: [967.638578208722, 424.0641291666773]
Distortion model: radtan
Distortion coefficients: [0.03459723604219442, -0.030767694481543133, -0.002550466204814176, 0.004007846658771188]
Type: aprilgrid
Tags:
Rows: 3
Cols: 4
Size: 0.046 [m]
Spacing 0.0138 [m]

IMU configuration

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IMU0:

Model: calibrated
Update rate: 200.0

Accelerometer:

Noise density: 0.000519

Noise density (discrete): 0.007339768388716364

Random walk: 4.62e-05

Gyroscope:

Noise density: 5.48e-05

Noise density (discrete): 0.000774989032180456

Random walk: 7.36e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

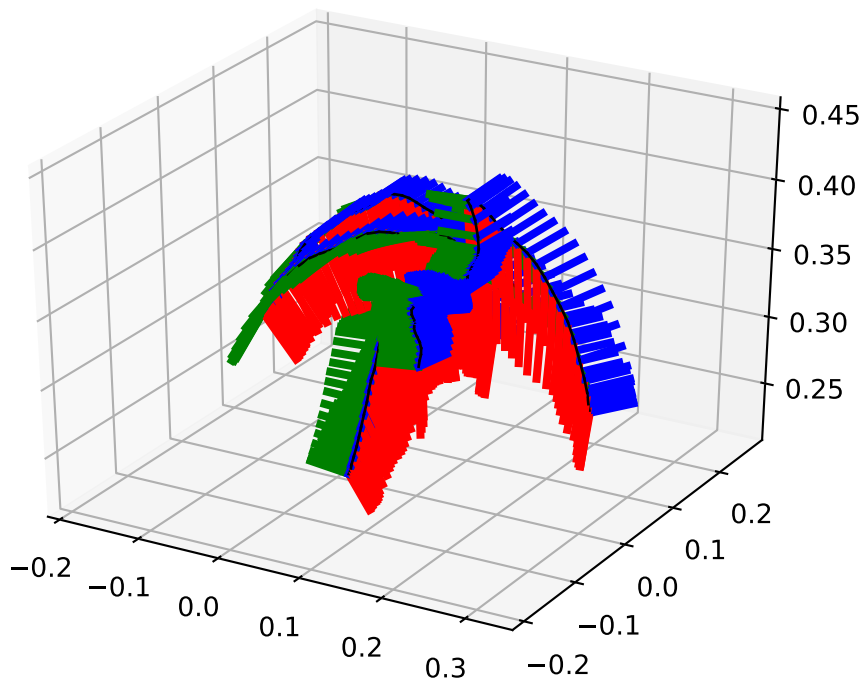
[0. 1. 0. 0.]

[0. 0. 1. 0.]

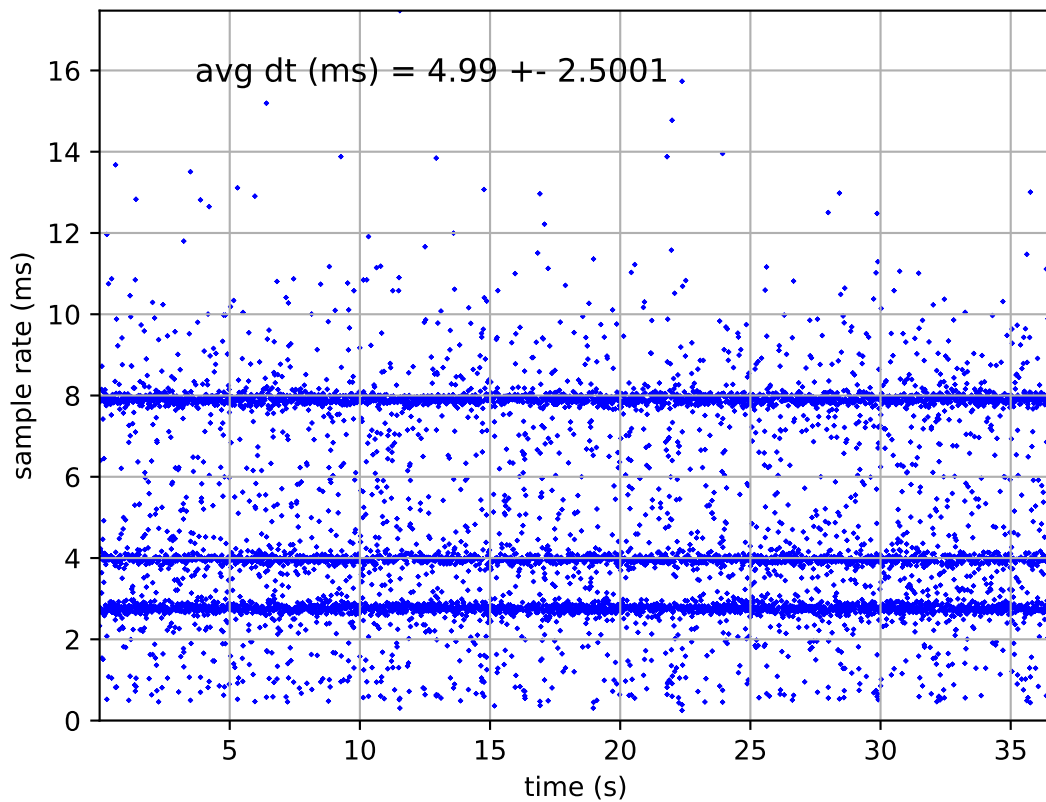
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

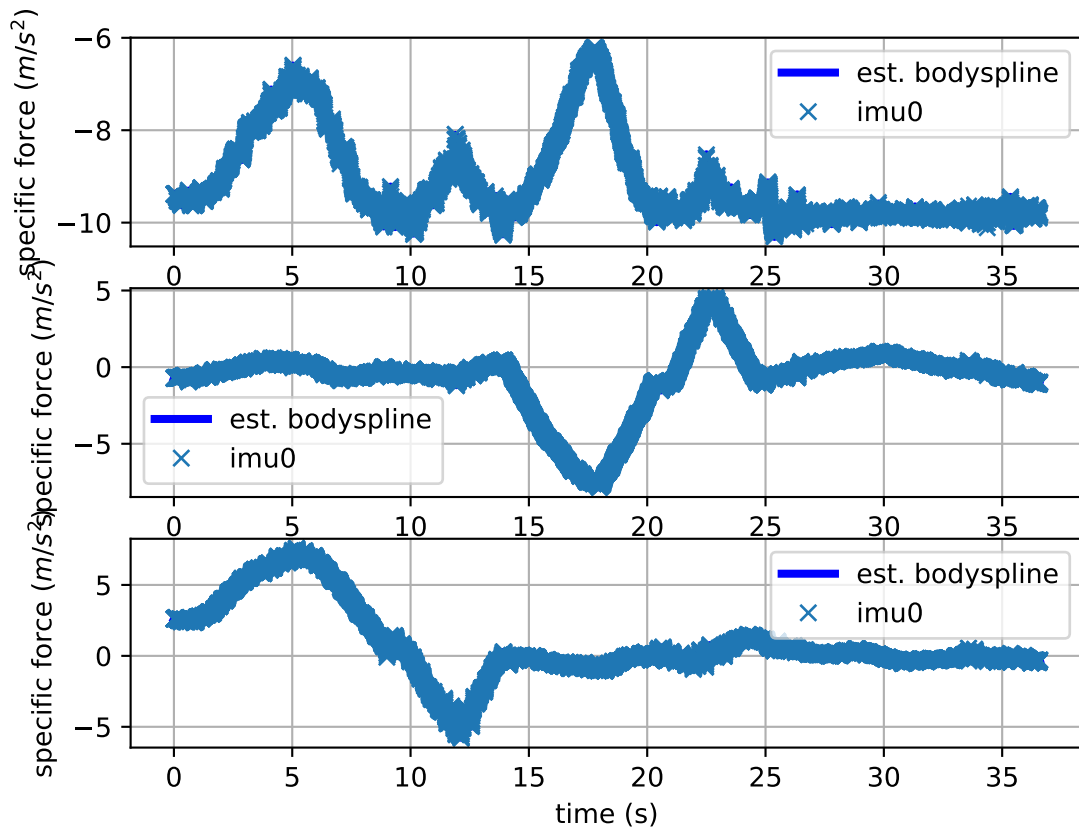
imu0: estimated poses



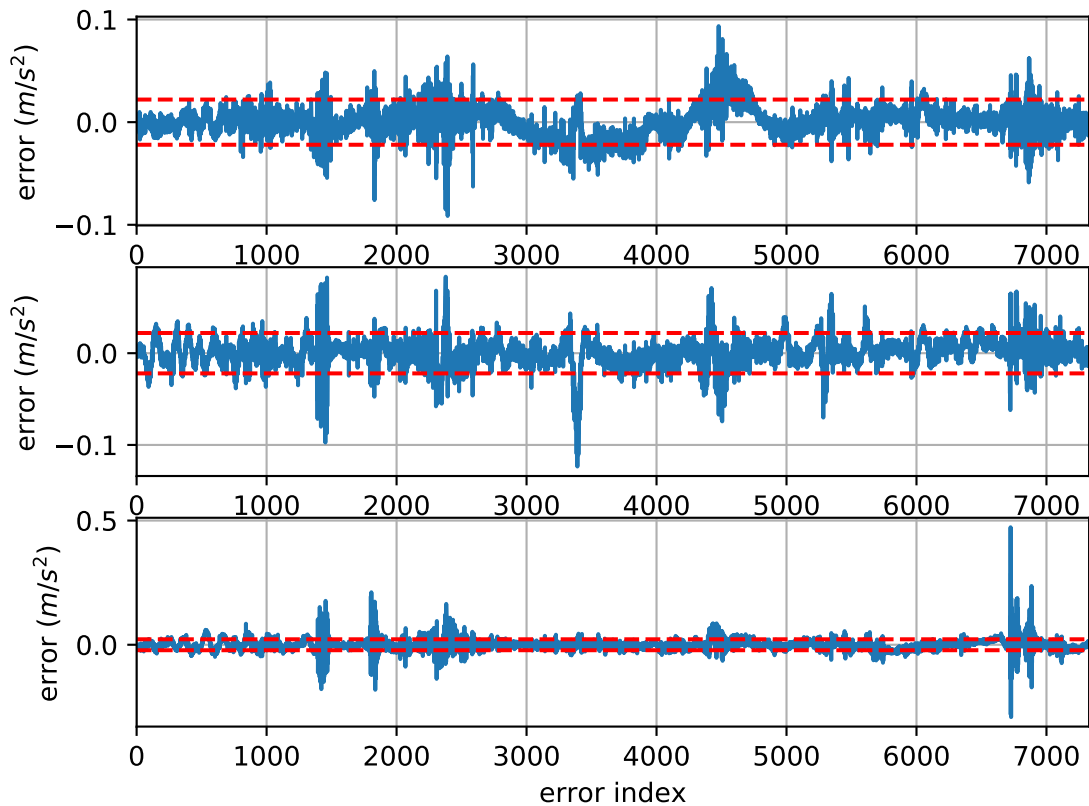
imu0: sample inertial rate



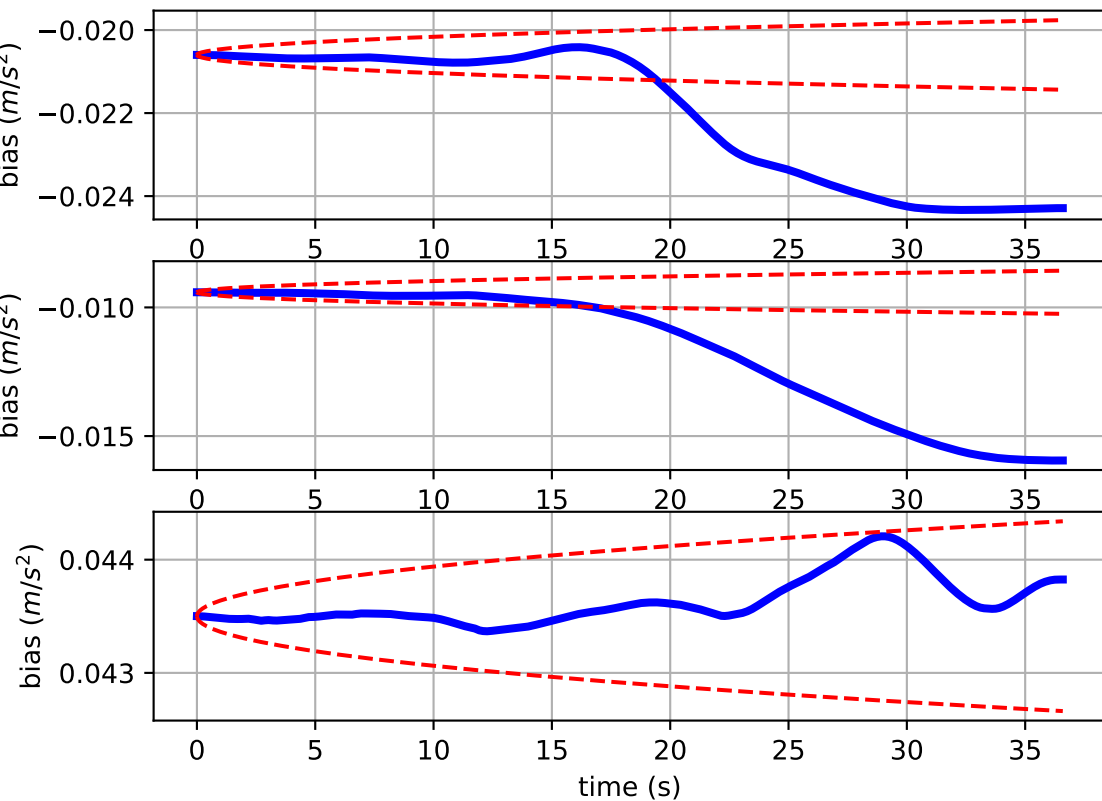
Comparison of predicted and measured specific force (imu0 frame)



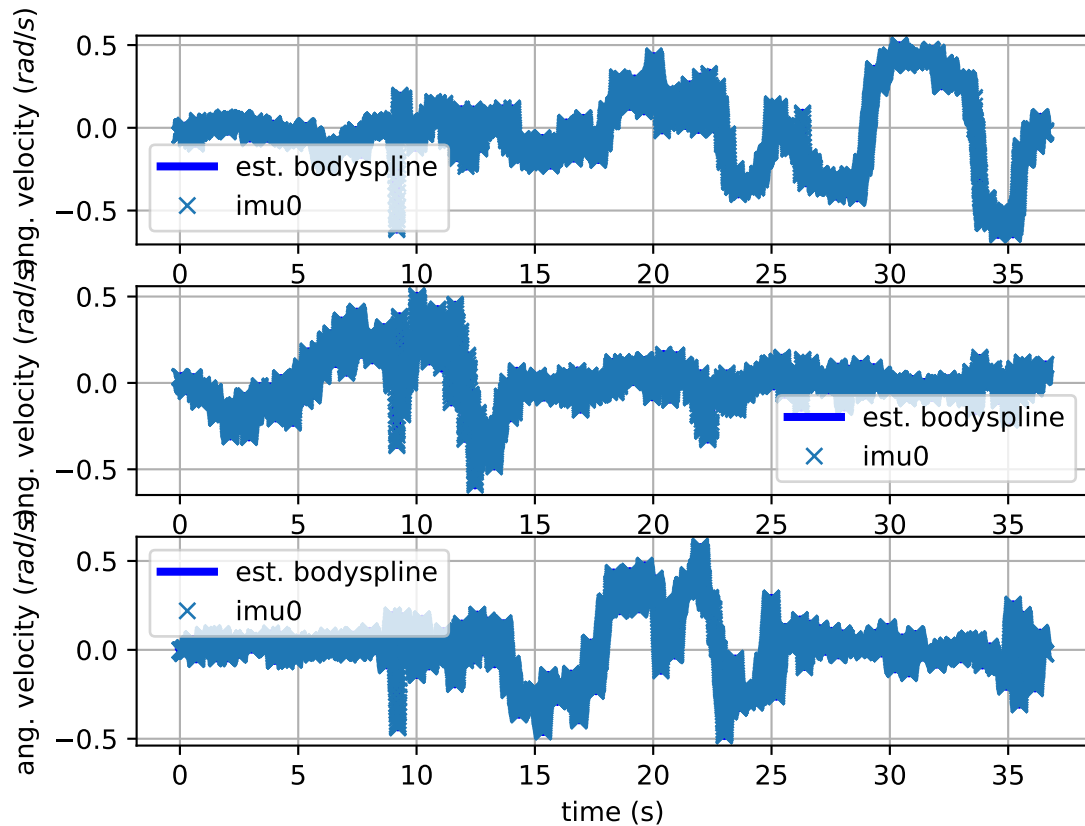
imu0: acceleration error



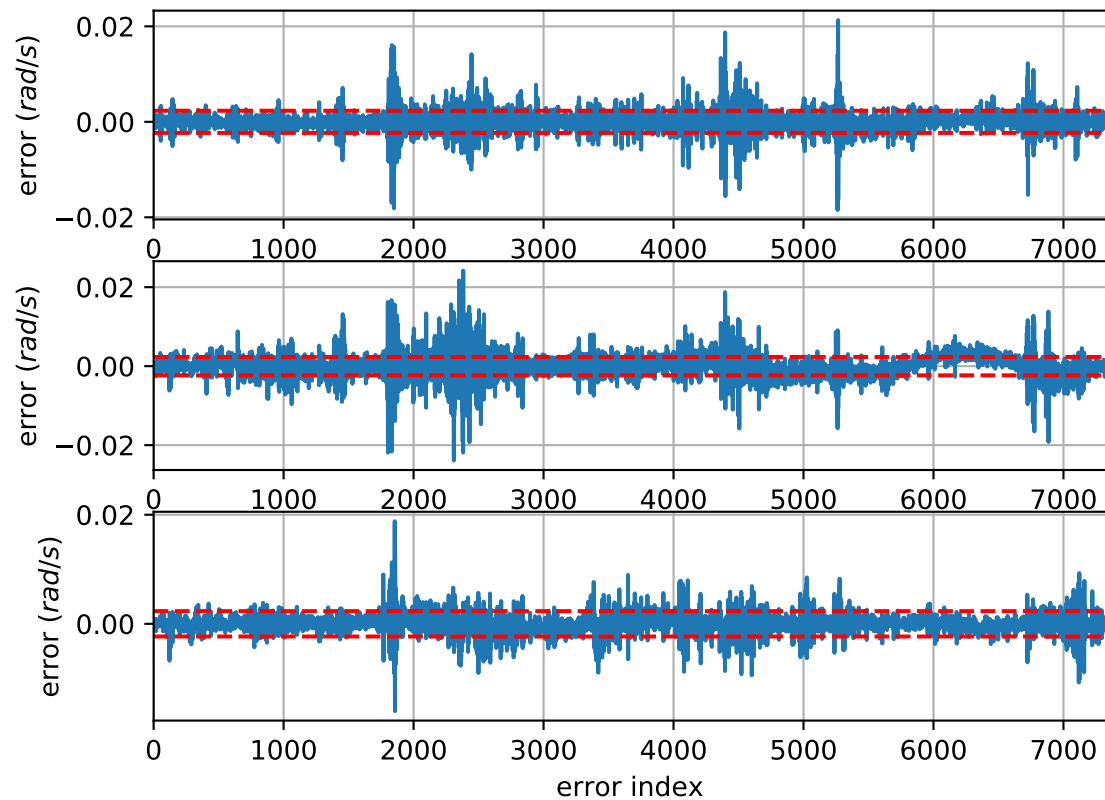
imu0: estimated accelerometer bias (imu frame)



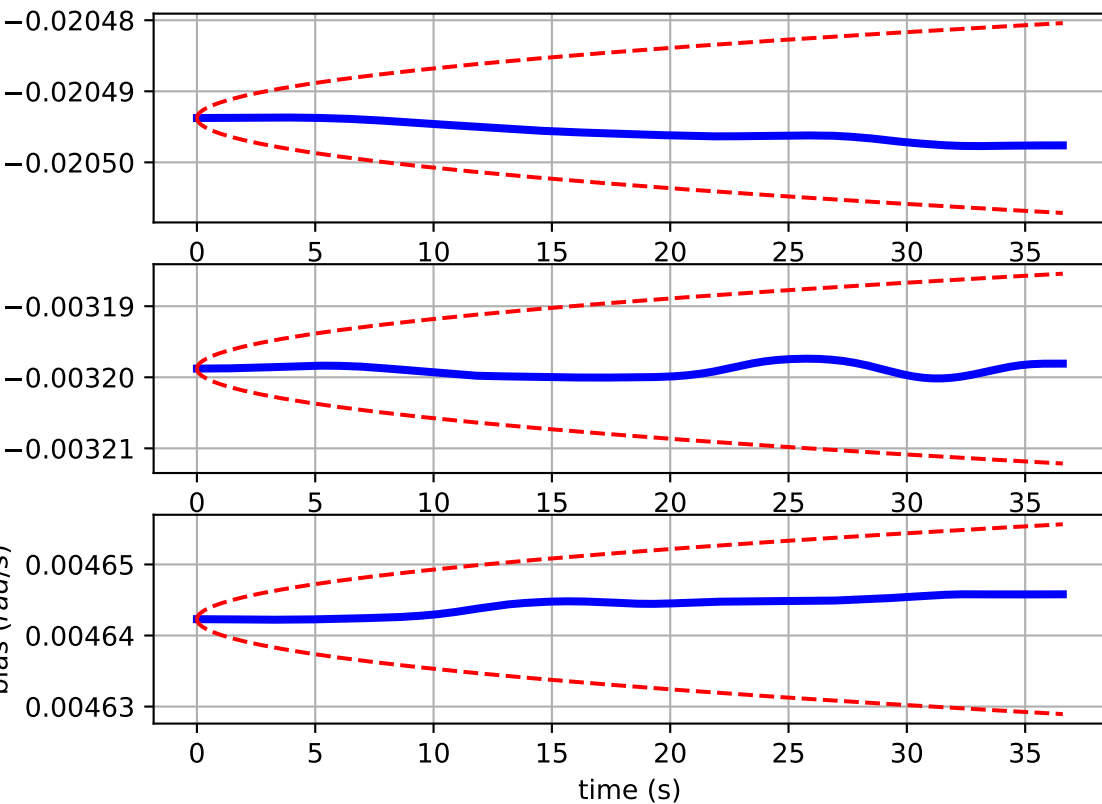
Comparison of predicted and measured angular velocities (body frame)



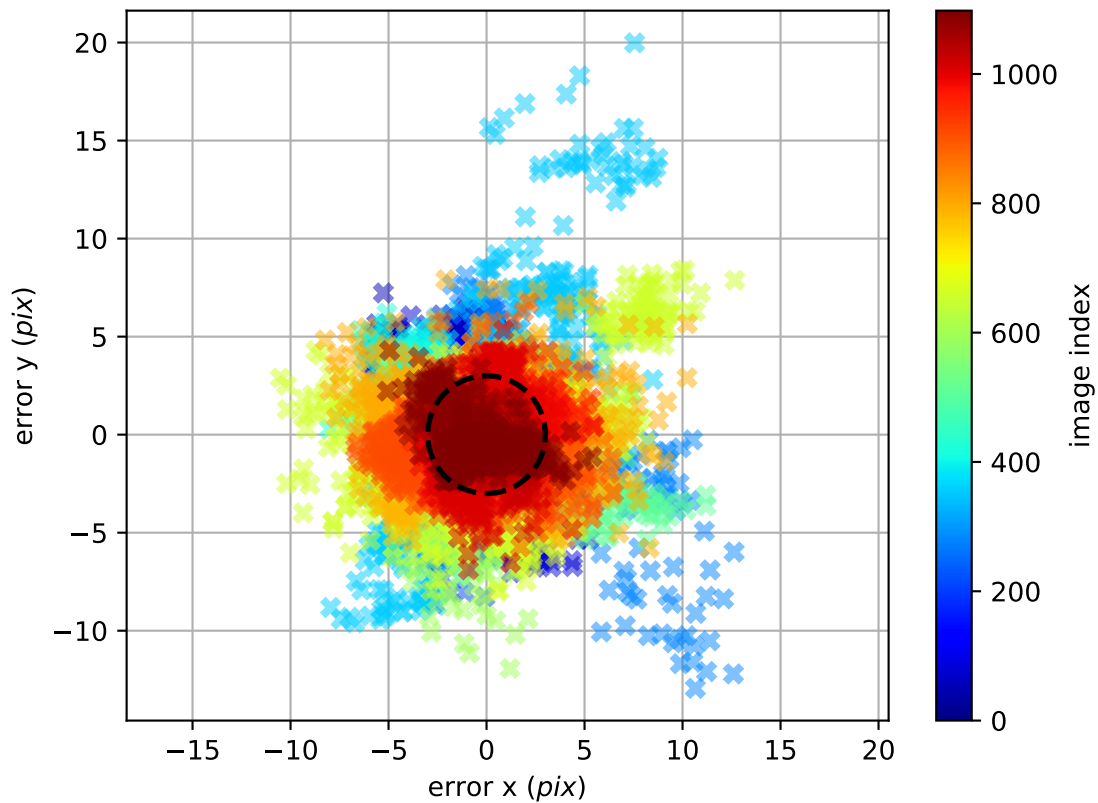
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

